

Data Poisoning and Dataset Trust

Can We Trust the Evidence?

**DATA
TRUST**

Today we use data poisoning as a practical way to check the sources, datasets, and claims behind your final project.

Due May 26, 2026

Plain-language definition

Data poisoning means bad inputs can shape bad outputs

Bad data can train bad decisions.

- Wrong labels can teach the wrong pattern
- Missing data can hide harm
- Manipulated data can make a system trust the wrong signal

Why this matters

Evidence is never neutral just because it is in a table

Source

Who collected the data and why?

Structure

What does one row mean, and what do the columns leave out?

Use

What claim are you trying to support with it?

Three easy failure modes

How final project evidence can mislead you

- The dataset is incomplete: important cases are missing
- The labels are unclear: categories hide important differences
- The source is one-sided: the evidence supports only part of the story

Final project use

Your project needs an evidence plan

1

Name the evidence you will use.

2

Explain where it came from.

3

Say what it can show.

4

Say what it cannot prove.

5

Use that limit to avoid overclaiming.

Coding path

For coding projects: inspect before you calculate

- What does one row represent?
- Which columns are signals, and which are context?
- Are there missing, unclear, or coded values?
- Could your score or chart overstate what happened?

Essay path

For MLA essays: sources also need trust checks

- Who wrote it, and what is their purpose?
- What evidence does the source use?
- What does the source leave out?
- What other source could balance it?

Privacy and ethics

Trust also means asking who could be harmed

- Does the evidence describe people directly or indirectly?
- Could location, time, or categories expose patterns?
- Could the data be used to judge a group unfairly?
- Could missing data make the conclusion unfair?

Today's lab

Lab goal: make your evidence safer to use

- Choose your project path
- List your evidence
- Check your strongest source or dataset
- Explain one way it could mislead you
- Write an evidence plan

Exit writing

Before you leave: name the risk of overclaiming

What could make my evidence misleading?

- Name one evidence item
- Name one possible weakness
- Name one step you will take
- Connect to security, privacy, or ethics

Source anchors

How today connects to security practice

- NIST frames adversarial machine learning as a real risk area
- OWASP identifies data poisoning as a machine learning security risk
- Your project uses the same habit: question the evidence before trusting the output

Work time

Use the rest of class to build your project

- Open the lab handout
- Work individually
- Ask for help narrowing your question
- Submit the completed lab by the end of class